Name of Institute: IITE Indus University Ahmedabad
Name of Subject Coordinator: Asst. Prof. Dipali Jitiya

Course code: CE0418
Course name: Operating System
Pre-requisites: Basic Programming, Data Structure
Credit points: 4
Offered Semester: IV

Course coordinator
Full name: Ms. Dipali Panchal
Department with sitting location: Bhawar Building,3rd Floor, Faculty Staffroom
Email: dipalipanchal.ce@indusuni.ac.in
Consultation times: Thursday: 3:00 PM to 4:00 PM

Students will be contacted throughout the Session via Mail with important information relating to this Course.

Course Objectives

1. To understand the services provided by and the design of an operating system.
2. To understand the structure and organization of the file system.
3. To understand what a process is and how processes are synchronized and scheduled.
4. To understand different approaches to memory management.
5. Students should be able to use system calls for managing processes, memory and the file system.
6. Students should understand the data structures and algorithms used to implement an

Course Outcomes (CO)

At the end of this subject, students should be able to:

1. Describe the general architecture of computers. [BT-2]
2. Describe, contrast and compare differing structures for operating systems. [BT-4]
3. Understand and analyze theory and implementation of processes, resource control (concurrency etc.), physical and virtual memory, scheduling, I/O and files. [BT-2,4]
4. Run Linux commands. [BT-6]
5. Understand background control command. [BT-2]
6. Understand process scheduling in program. [BT-3]

Course Outline

Processor management, multiplexing, interrupts, multiprocessing, Memory management, partitions, swapping, paging, disks, files, directories, Input/Output, buffering. Job scheduling. Networks. Case study.

Method of delivery

Face to Face lectures, self-study material, Active Learning Techniques

Study time

3 Hours/week

CONTENTS

UNIT-I **[12 hours]**

Introduction to Operating System

Architecture of OS (Ex. Monolithic, Microkernel, Layered , Exo kernel), Operating system objectives and functions, Virtual Computers, Interaction of O. S. & hardware architecture, Evolution of operating systems, Batch, multiprogramming. Multitasking, Multiuser, parallel, distributed & real – time O.S., System calls, O. S. Shell, Linux Shell commands, Shell programming. Examples of O. S.: Linux, MS- Windows, Handheld OS.

Process Management

Process, Process description, Process states, Process control, Threads, Processes and Threads, Uniprocessor Scheduling: Types of scheduling, Scheduling algorithms: FCFS, SJF, Priority, Round Robin, UNIX Multi-level feedback queue scheduling, Thread Scheduling, Multiprocessor Scheduling concept, Real Time Scheduling concept

UNIT-II **[12 hours]**

Concurrency

Principles of Concurrency, Critical Section problem, Mutual Exclusion H/W Support, software approaches, Semaphores and Mutex, Message Passing, Monitors, Classical Problems of **Synchronization:**

Readers-Writers Problem, Producer Consumer Problem, Dining Philosopher problem

Deadlock:

System model, Deadlock characterization, Methods for handling deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from deadlock.

UNIT-III **[12 hours]**

Memory Management

Memory Management requirements, Memory partitioning: Fixed ,dynamic, partitioning, Contiguous memory allocation, Buddy System Memory allocation Strategies (First Fit, Best Fit, Worst Fit, Next Fit), Fragmentation, Swapping, Segmentation , Paging, Virtual Memory, Demand paging, Page Replacement Policies (FIFO, LRU, Optimal, clock) ,Thrashing.

I/O Management and Disk Scheduling

I/O Devices, Organization of I/O functions, Operating System Design issues, I/O Buffering, Disk Scheduling (FCFS, SCAN, C-SCAN, SSTF), Disk Caches.

UNIT-IV **[12 hours]**

File Management

Overview, File Organization and access, File Directories, File Sharing, Security issues, Record Blocking, Secondary Storage Management.

Computer security & protection:
Security Threats, Attacks and assets, Intruders, Malicious software, Protection: Protection Policy and mechanisms, Authentications: Internal Access Authorizations, Implementations.

CO-PO Mapping (PO: Program Outcomes)

1 Program Outcomes (PO's)

Engineering Graduates will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6 The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

2. Programme Specific Outcome

PSO1. To assess the hardware and software aspects of computer systems.
PSO2. To analyze and develop difference packages working with specific hardware systems.

COURSE OUTCOME (CO) and PROGRAM OUTCOME (PO) Matrix
(1- Low, 2-Medium, 3- High)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C0 1	3	2	-	2	1	-	-	-	-	-	-	-
C0 2	3	2	1	1	-	-	-	-	-	-	-	2
C0 3	3	2	3	2	2	-	-	1	1	1	-	2
C0 4	2	1	2	-	-	-	-	-	-	-	-	2
C0 5	2	1	1	-	-	-	-	-	-	-	-	2
C0 6	2	2	2	1	-	-	-	-	-	1	-	1
CE0418	**2.5**	**1.7**	**1.5**	**1.5**	**1.5**	**0.0**	**0.0**	**1.0**	**1.0**	**1.0**	**0.0**	**1.8**

Average = (total of all COs for each PO)/(no of entries in that PO)

COURSE OUTCOME and PROGRAM SPECIFIC OUTCOME Matrix

Operating system (CE0418)

CO	PSO1	PSO2
CO 1	2	2
CO2	2	2
CO 3	3	3
CO 4	1	-
CO 5	1	1
CO 6	1	1
	1.7	**1.5**

Blooms Taxonomy and Knowledge retention (For reference)

(Blooms taxonomy has been given for reference)

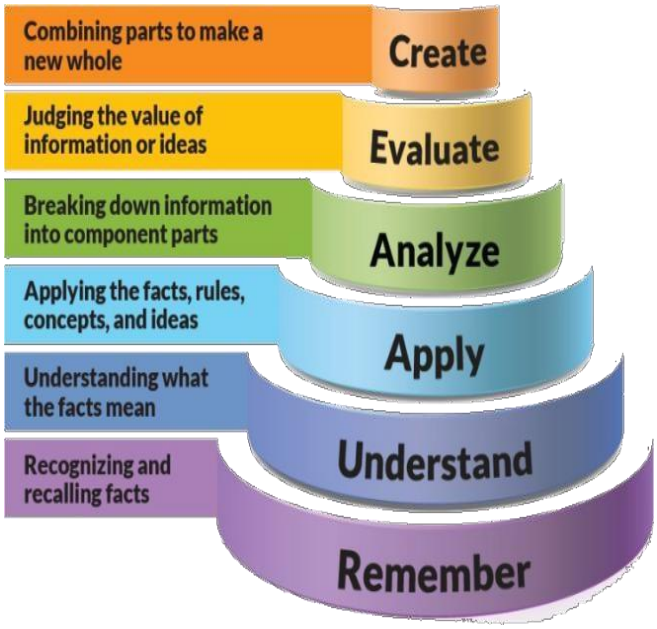


Figure 1: Blooms Taxonomy

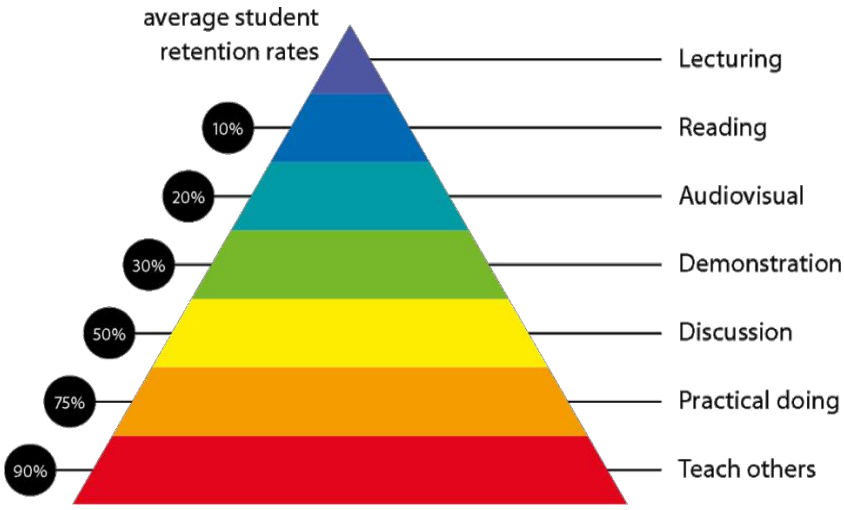


Figure 2: Knowledge retention

Practical work:

Wk No.	Class Activity	List of Practical
01	Lab 1	Study Linux/UNIX Commands. 1. Directory Commands : mkdir, rmdir, cd, pwd, ls, mv, help, man, chmod, etc. 2. File or Text handling Commands : touch, cp, mv, rm, sort, cat, pg, lp, pr, file, find, more, cmp,diff, comm, head, tail, cut, grep, etc. 3. Information Commands : learn, man, who, date, cal, tty, calendar, time, bc, whoami, which, hostname, history, etc.
02	Lab 2	2.1 Write a shell script to generate mark sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student. 2.2 Write a shell script to input two no's from the user and perform addition, subtraction, multiplication, and division.
03	Lab 3	3.1 Write a shell script that determines whether the entered year is a leap year or not. 3.2 Write a shell script to display first n prime numbers as output.
04	Lab 4	4.1 Write a shell script to find factorial of given number n. 4.2 Write a shell script which will generate first n Fibonacci numbers.
05	Lab 5	5.1 Write a shell script to check entered string is palindrome or not. 5.2 Write a shell script to merge content of two text files.
06	Lab 6	6.1 Write a C program for implementation of FCFS scheduling algorithms 6.2 Write a C program for implementation of SJF scheduling algorithms
07	Lab 7	7.1 Write a program for implementation of Round Robin scheduling algorithms.
08	Lab 8	Process control system calls: The demonstration of fork() along with zombie and orphan states.

Lecture/tutorial times

As per timetable provided by department.

The University norms states that it is the responsibility of students to attend all lectures, tutorials, seminars and practical work as stipulated in the Course outline. Minimum attendance requirement as per university norms is compulsory for being eligible for mid and end semester examinations.

Details of referencing system to be used in written work

Text books:

1. Silberschatz, Galvin, and Gagne, "Operating System Concepts", By, Wiley India Pvt Ltd.
2. William Stallings, "Operating Systems Internals and Design Principles", By, Pearson Education.

Reference Books:

1. Andrew S. Tanenbaum, "Modern Operating Systems", By, Pearson Education.
2. Gary Nutt, "Operating Systems", By, Pearson Education.
3. Harvey M. Deital, "Operating Systems", By, Pearson Education.
4. Das Sumitabha," Unix Concepts and Applications", Tata McGraw Hill.

Additional Materials:

1) http://www.nptel.ac.in/
2) https://www.tutorialpoint.com/os/

ASSESSMENT GUIDELINES

Theory:	***Practical:***
CIE-TH (60)	**CIE-PR (60)**
Mid Semester Exam [40 Marks]	Practical Performance, Regularity –[20 Marks]
Assignment [10 Marks]	Practical Exam (Internal) – [20 Marks]
Regularity [05 Marks]	practical File Submission-[10 Marks]
Viva – [05 Marks]	Viva- [10 Marks]
ESE-TH (40)	**ESE-PR (40)**
Total: 200 Marks	

Your final course mark will be calculated from the following:

SUPPLEMENTARY ASSESSMENT

Students who receive an overall mark less than 40% in mid semester or end semester will be considered for supplementary assessment in the respective components (i.e mid semester or end semester) of semester concerned. Students must make themselves available during the supplementary examination period to take up the respective components (mid semester or end semester) and need to obtain the required minimum 40% marks to clear the concerned components.

Practical Work Report/Laboratory Report:

A report on the practical work is due the subsequent week after completion of the class by each group.

Late Work

Late assignments will not be accepted without supporting documentation. Late submission of the reports will result in a deduction of -% of the maximum mark per calendar day

Format

All assignments must be presented in a neat, legible format with all information sources correctly referenced. **Assignment material handed in throughout the session that is not neat and legible will not be marked and will be returned to the student.**

Retention of Written Work

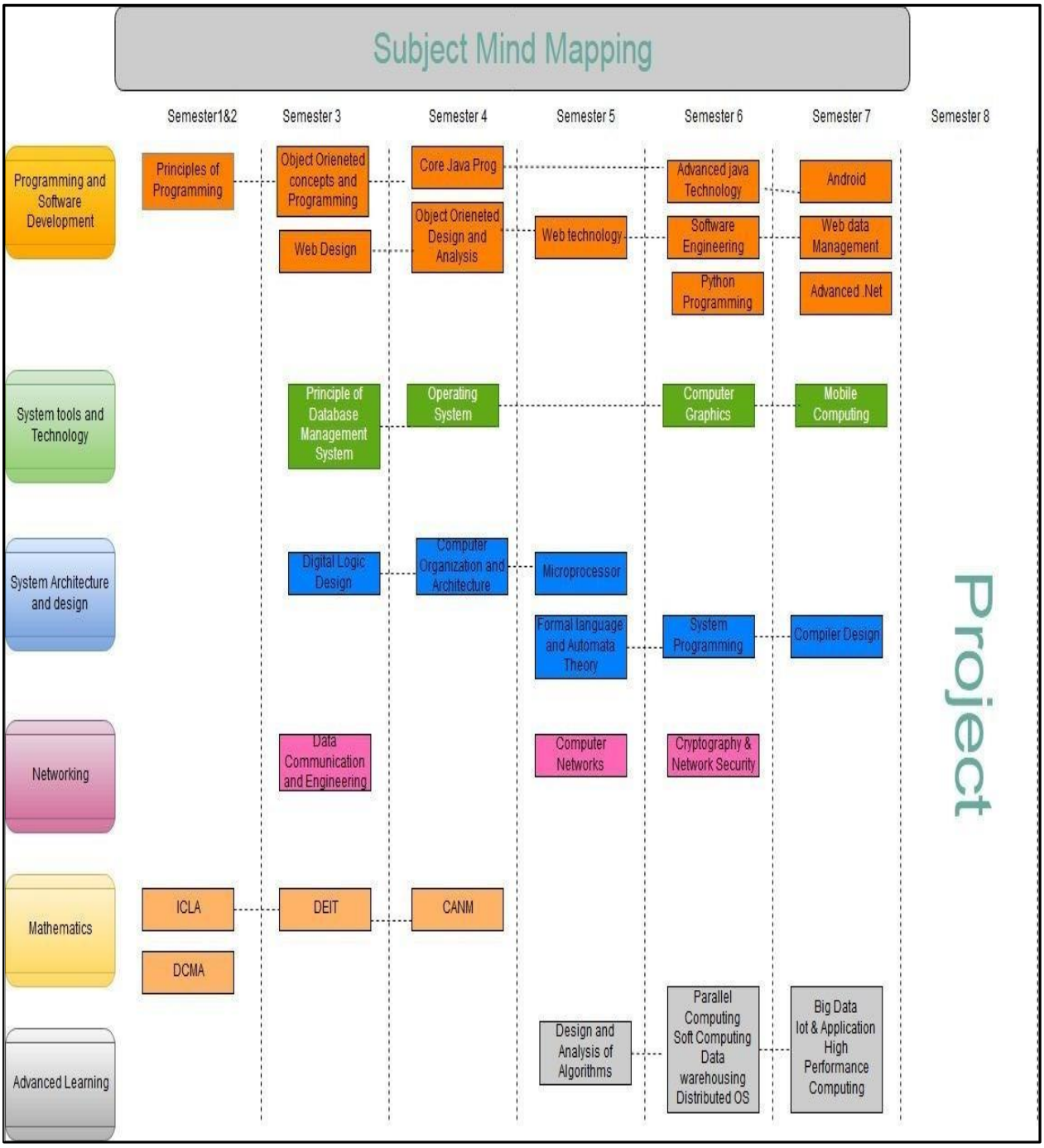
Written assessment work will be retained by the Course coordinator/lecturer for two weeks after marking to be collected by the students.

University and Faculty Policies

Students should make themselves aware of the University and/or Faculty Policies regarding plagiarism, special consideration, supplementary examinations and other educational issues and student matters.

Plagiarism - Plagiarism is not acceptable and may result in the imposition of severe penalties. Plagiarism is the use of another person's work, or idea, as if it is his or her own - if you have any doubts at all on what constitutes plagiarism, please consult your Course coordinator or lecturer. Plagiarism will be penalized severely.

Do not copy the work of other students.
Do not share your work with other students (except where required for a group activity or assessment) Course schedule (subject to change)

(Mention quiz, assignment submission, breaks etc as well in the table under the Teaching Learning Activity Column)

PROGRAM MAP for Bachelor of Engineering
(CE / CSE / IT)